PassFrame tuition

Factors, Multiples & Primes

Year 4 Pt 2 — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 1, 2, 4, 8

2. 1, 3, 5, 15

3. A, C and E (12, 16 and 20 are multiples of 4; 14 and 18 are not)

4. 6, 12, 18, 24, 30

5. Yes

6. No

7. A — Yes

8. No — it has factors other than 1 and itself

9. 1, 2, 4, 5, 10, 20

10. B — 45

11. 11, 13, 17, 19

12. Yes

13. B — 4

14. 15

15. B — No, prime numbers have exactly two factors, but 1 has only one factor

Standard

1. 1, 2, 3, 6

2. C — 8

3. 12

4. 24

5. 1×36, 2×18, 3×12, 4×9, 6×6

6. B — No, 51 = 3 × 17, so it has more than two factors

7. 10, 20, 30

8. 10

9. A, C and E (36, 49 and 64 are square numbers; 45 and 60 are not)

10. Yes — 8 × 8 = 64

11. 1, 2, 3, 4, 6, 8, 12, 16, 24, 48

12. A and C (23 and 29 are prime; 27 = 3×9 and 33 = 3×11 are not)

13. 6

14. C — 12

15. e.g. 7 + 23, 11 + 19 or 13 + 17 (any valid pair)

Standard / Hard

1. 24

2. B — 53 (5+3=8, and 53 is prime)

3. 61

4. 18

5. A, C and D (10, 20 and 30 are all factors of 60 and multiples of 10; 15 is not a multiple of 10, and 40 is not a factor of 60)

6. 31

7. 60

8. B — 8 and 12 (HCF(8,12)=4)

9. 1, 4, 25, 100

10. 28

11. A — 2 × 3 × 5 = 30

12. e.g. 6, 8, 10, 14 or 15 (any number with exactly 4 factors)

13. e.g. 3, 6, 12 or 24 (any number whose LCM with 8 is 24)

14. A and B (3 and 5 are prime factors of 45; 9, 15 and 45 are factors but not prime)

15. 1×48, 2×24, 3×16, 4×12, 6×8 (cm)

Homework — Answers

Easy / Standard

1. 1, 2, 5, 10

2. 1, 2, 4, 8, 16

3. A, C and E (15, 20 and 25 are multiples of 5; 18 and 24 are not)

4. 7, 14, 21, 28, 35

5. Yes

6. No

7. A — Yes

8. No

9. 1, 2, 3, 4, 6, 8, 12, 24

10. B — 54

11. 23, 29

12. Yes

13. A — 4

14. 12

15. A — Yes

Standard

1. 1, 2, 5, 10

2. C — 9

3. 30

4. 36

5. 1×40, 2×20, 4×10, 5×8

6. B — No, 57 = 3 × 19

7. 12, 24, 36

8. 5

9. A, C and E (100, 121 and 144 are square numbers; 110 and 130 are not)

10. Yes — 11 × 11 = 121

11. 1, 2, 3, 6, 9, 18, 27, 54

12. A and D (41 and 53 are prime; 45 = 9×5 and 49 = 7×7 are not)

13. 4

14. C — 15

15. e.g. 5 + 19, 7 + 17 or 11 + 13 (any valid pair)

Standard / Advanced

1. 2, 4, 6, 12, 18, 36

2. 24

3. 12; the sum of two odd primes is always even

4. B — 22 (LCM=24, HCF=2, 24−2=22)

5. 11

6. 12

7. C — 24 (LCM of 6 and 8 is 24)

8. 42

9. 27

10. C — 60 (LCM of 4, 6 and 10 is 60)

11. 4, 16, 36, 64

12. 15 pupils per group; 11 groups in total

13. A, B and D (101 is the number: it is odd and prime, and one more than 100 which is a multiple of 10; it is not itself a multiple of 10, and 100 is not a factor of a prime number)

14. 45

15. A — width 2 m, length 36 m (2 is prime and 2×36 = 72; the other options have widths 4, 6 or 8, none of which are prime)